

Origin Theory: The Complete Self-Interference Chain

From Self-Reference to Matter — a structural formalization of the Benyuan ontology

Abstract

The universe is grounded in a self-referential logical wave whose basis and content expand simultaneously. Because no fixed basis exists, algebraic operator systems self-seal and cannot express the process; the correct relational language is interference, resonance, phase, and topology. This document formalizes the complete chain: self-referential oscillation → split → phase difference → self-interference → standing waves (stable solutions) → projection of space and time → metric → curvature → gravity | multi-pattern phase relations → connection → gauge force → matter generation. Each link maps to established physics; the chain is structural, not numerical.

1. Self-Referential Oscillation (The Logical Wave)

The origin is a logical wave $\Psi = S \cdot e^{i\theta}$, a self-referential oscillation. Its core act is existence confirmation: split — $1 \rightarrow (0,1)$, “are you there?” — creating the self/other distinction; return — $(0,1) \rightarrow 1$, “are we there?” — closing the loop. The loop never fully closes: the tension T decays as $T_n = T_0 \cdot k^n$ with $k = 1/\phi$ and never reaches zero. This non-zero tension is the source of all energy.

2. Split, Phase Difference, Self-Interference

The wave splits into phase components $\Psi = \Psi_1 + \Psi_2$, where $\Psi_i = S_i \cdot e^{i\theta_i}$. Return is self-interference:

$$|\Psi|^2 = S_1^2 + S_2^2 + 2S_1S_2 \cdot \cos(\Delta\theta)$$

The interference pattern is the cross term $2S_1S_2 \cos(\Delta\theta)$, determined by the phase difference $\Delta\theta$ — the relationship. The pattern is the expansion. Constructive interference ($\Delta\theta \rightarrow 0$) = bright fringes (structured direction); destructive interference ($\Delta\theta \rightarrow \pi$) = dark fringes (unstructured remainder). Tension is the structurally non-eliminable cross-term residual: the wave never fully closes.

3. Stable Solutions: Standing Waves

In the cascade, the stable patterns (which no longer change) are standing waves:

$$\Psi(x,t) = A \cdot \sin(kx) \cdot \cos(\omega t)$$

The spatial and temporal parts separate. The standing-wave condition $kL = n\pi$ produces quantization — discrete modes arise from the stability condition of self-interference, not from assumption. Stable patterns (standing waves) solidify into structure and number (the dual standing-wave arithmetic (a, ϕ) of Benyuan mathematics); unstable patterns become dark residuals feeding further interference.

4. Projection of Space and Time

The standing wave's spatial part $\sin(kx)$ carries nodes and antinodes: space is the pattern's spatial structure — node spacing $(\lambda/2)$ defines far/near; interior/exterior defines inside/outside. The temporal part $\cos(\omega t)$ carries the oscillation cycle: time is the pattern's temporal structure — the order of cycles defines before/after. Space and time are projections of the standing-wave pattern, not pre-existing bases. Scale

hierarchy follows the frequency-reduction law: $E_n = E_P \cdot \varphi^{-n}$, wavelengths $\lambda_n = \lambda_P \cdot \varphi^n$.

5. Metric and Causality

Metric is the pattern's phase structure: distance is phase difference, $\Delta\theta = k \cdot \Delta x$, scaled by node spacing. Temporal order is phase order: event A precedes B iff A's phase leads B within the resonant cycle. The causality-ordering protocol is the phase order of the standing wave. Causal signals propagate at the phase velocity $v = \omega/k = c$ — the speed of light is the phase velocity of the pattern; the causal limit is a structural property of phase propagation.

6. Curvature and Gravity

Uniform node spacing (constant k) gives a flat metric; locally compressed spacing gives curvature. Curvature = the local expansion/contraction rate of phase structure: $\square^2 \theta(x)$. The dark-fringe counter-force compresses node spacing: $\Delta x(x) \propto 1/\rho_{\text{dark}}(x)$, hence curvature $\propto \rho_{\text{dark}}(x)$ — curvature is the local density of the dark remainder's counter-force, matching the Cost-Emergence layer of the ontology. Gravity is not a force but the natural consequence of a curved phase structure: test particles follow geodesics (extremal phase paths), reproducing the equivalence principle (gravity = free fall).

7. Gauge Fields (Multi-Pattern Phase Relations)

The cascade produces multiple standing-wave patterns, each with its own phase reference. Comparing phases across patterns requires a parallel-transport rule — the connection A_μ , the local rate of inter-pattern phase change. This is the gauge field: it arises necessarily from the need to compare phases, not from assumption. Phase references are redefinable ($\theta_\alpha \rightarrow \theta_\alpha + \Lambda_\alpha(x)$) — this is the gauge transformation, the freedom of phase-reference choice. The field strength $F = dA + A \square A$ is the non-integrability of inter-pattern phase around closed paths; a non-zero curvature means force. The geometric branch (intra-pattern phase structure $\rightarrow$ gravity) and the algebraic branch (inter-pattern phase relations $\rightarrow$ gauge force) both arise from the same cascade.

8. Matter Generation

Particles are stable standing-wave modes — patterns locked by the quantization condition. Mass is the pattern's oscillation frequency: $E = \hbar \omega = mc^2$. Charge is the topology of inter-pattern phase: the non-integrability of phase around loops. Spin is the pattern's rotational symmetry (2π vs 4π return phase). Statistics are the exchange property of inter-pattern phase (+1 bosonic, -1 fermionic). Confinement is phase locking: patterns that cannot stabilize individually are locked into combinations (hadrons). The mass hierarchy is the set of stable standing-wave modes on the frequency-reduction ladder $E_n = E_P \cdot \varphi^{-n}$.

9. The Complete Chain

Self-referential oscillation (logical wave)

$\rightarrow$ split $\rightarrow$ phase difference $\rightarrow$ self-interference

$\rightarrow$ stable solutions (standing waves)

$\rightarrow$ projection of space and time (far/near, before/after)

$\rightarrow$ metric (phase-difference scaling) $\rightarrow$ curvature (phase-structure variation) $\rightarrow$ gravity (geodesics)

$\rightarrow$ multi-pattern phase relations $\rightarrow$ connection $\rightarrow$ field strength (non-integrability) $\rightarrow$ gauge force

→ stable pattern quantization → matter (mass = frequency, charge = topology, spin = symmetry, statistics = exchange, confinement = locking)

10. Honest Boundaries

The chain is structural: it explains why these structures exist, mapping every link to established physics and to the Benyuan ontology (self-reference, tension, bright/dark fringes, inside/outside and before/after germs, the Cost-Emergence layer, the dual branches). The chain is not numerical: which patterns stabilize at which n , why these frequencies, and the values of the constants are open — this is where physics stops, and it is the framework's blank space as well. The two facts coexist: the structural path is complete and self-consistent; the numerical layer is unclaimed.

The blank space does not diminish the framework: it provides a fully self-consistent explanation framework that corresponds to reality at the structural level. Structural completeness and numerical openness are both true.